

A BOUNDARY-CONSTANT OBSTRUCTION TO REAL-ROOTED RIEMANN INCOMPLETE-GAMMA APPROXIMANTS

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ABSTRACT. Let H_N denote the N th Fourier partial sum obtained from Riemann's incomplete-gamma expansion, and let

$$c_N = \sum_{n>N} (4\pi n^2 - 1)e^{-\pi n^2} > 0.$$

For every finite $N \geq 1$ and every $0 < t \leq 1$, we prove that the even entire function $F_{N,t} = H_N + tc_N$ has a nonreal zero. The direct arithmetic-index truncation is the endpoint $F_{N,1}$. Indeed, $F_{N,t}(x) \rightarrow tc_N$ at both ends of the real axis, while $F_{N,t}(i/2) - tc_N = 1/2 - c_N \neq 0$; moreover, $F_{N,t}$ has order at most one. If all zeros were real, the positive boundary limit would leave only finitely many zeros, and Hadamard factorization together with evenness would force $F_{N,t}$ to be constant. Pólya's theorem already treats the unshifted endpoint $t = 0$. The result identifies the obstruction created by the positive constant shift and separates it from Haglund's next-summand zero-motion conjectures. It has no implication for the Riemann hypothesis itself.

1. INTRODUCTION

To support a limiting real-rootedness argument, a finite approximating family needs a corresponding finite-level zero theorem. Local uniform convergence alone does not place the approximants' zeros on the real axis. Pólya's Fourier-transform program and Hejhal's refinement show that finite-level zero behavior depends sensitively on the choice of approximating kernel [5, 4]. Haglund later studied incomplete-gamma partial sums and the geometry of their complex zeros [3].

This note studies direct finite truncations of the symmetrized incomplete-gamma expansion for the completed zeta function. These truncations preserve both functional-equation halves term by term and converge locally uniformly to Ξ . Conversion to a cosine transform separates an additional boundary constant. The exact decomposition derived below is $\Xi_N = H_N + c_N$. We place it in the one-parameter family $F_{N,t} = H_N + tc_N$ and study every positive shift $0 < t \leq 1$.

The boundary constant forces a nonreal zero at every finite level.

Theorem 1.1. *For every finite integer $N \geq 1$ and every $0 < t \leq 1$, the function $F_{N,t}$ defined in (8) has at least one nonreal zero. In particular, the symmetrized incomplete-gamma approximant $\Xi_N = F_{N,1}$ has a nonreal zero for every finite N .*

The argument does not locate a zero numerically. It combines four global facts: a positive real-axis boundary limit, nonconstancy, order at most one, and evenness. Section 2 derives the exact cosine representation and its endpoint constant. Section 3 evaluates the constant by Jacobi's theta transformation. The growth estimate and factorization argument appear in Sections 4 and 5. Section 6 identifies the unshifted endpoint covered

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by Pólya, distinguishes Haglund's deformation, and states what the theorem does and does not imply.

2. THE FINITE INCOMPLETE-GAMMA FAMILY

Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right).$$

The standard theta-function argument gives a symmetrized incomplete-gamma expansion for ξ ; see, for example, [2, Ch. 1] and [3]. Its direct truncation after the N th arithmetic term is

$$(1) \quad \xi_N(s) = \frac{1}{2} + \frac{s(s-1)}{2} \sum_{n=1}^N \left[(\pi n^2)^{-s/2} \Gamma\left(\frac{s}{2}, \pi n^2\right) + (\pi n^2)^{-(1-s)/2} \Gamma\left(\frac{1-s}{2}, \pi n^2\right) \right],$$

where $\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t} dt$. Define

$$(2) \quad \Xi_N(z) = \xi_N\left(\frac{1}{2} + iz\right).$$

Each Ξ_N is a real even entire function. The omitted tail decays exponentially in n^2 , uniformly for z in compact sets, so $\Xi_N \rightarrow \Xi$ locally uniformly as $N \rightarrow \infty$.

Set

$$(3) \quad S_N(u) = \sum_{n=1}^N \exp\left(\frac{u}{2} - \pi n^2 e^{2u}\right), \quad u \geq 0,$$

and

$$I_N(z) = \int_0^\infty S_N(u) \cos(zu) du.$$

Substituting $t = \pi n^2 e^{2u}$ in the two incomplete-gamma integrals and using $s = 1/2 + iz$ gives

$$(4) \quad \Xi_N(z) = \frac{1}{2} - 2 \left(z^2 + \frac{1}{4}\right) I_N(z).$$

Let

$$(5) \quad \Phi_N(u) = S_N''(u) - \frac{1}{4} S_N(u).$$

All derivatives of S_N decay superexponentially at infinity. Two integrations by parts therefore yield

$$\int_0^\infty \Phi_N(u) \cos(zu) du = -S_N'(0) - \left(z^2 + \frac{1}{4}\right) I_N(z).$$

Combining this identity with (4) proves the exact representation

$$(6) \quad \Xi_N(z) = c_N + 2 \int_0^\infty \Phi_N(u) \cos(zu) du,$$

where

$$(7) \quad c_N = \frac{1}{2} + 2S_N'(0) = \frac{1}{2} + \sum_{n=1}^N (1 - 4\pi n^2) e^{-\pi n^2}.$$

Put

$$(8) \quad H_N(z) = 2 \int_0^\infty \Phi_N(u) \cos(zu) du, \quad F_{N,t}(z) = H_N(z) + t c_N \quad (0 \leq t \leq 1).$$

Then (6) says exactly that $\Xi_N = F_{N,1}$.

3. THE POSITIVE BOUNDARY CONSTANT

The sign of c_N follows from the theta transformation, not from numerical cancellation in (7).

Proposition 3.1. *For every finite $N \geq 1$,*

$$(9) \quad c_N = \sum_{n>N} (4\pi n^2 - 1)e^{-\pi n^2}, \quad 0 < c_N < \frac{1}{2}.$$

Moreover,

$$(10) \quad \lim_{x \rightarrow \pm\infty} \Xi_N(x) = c_N, \quad x \in \mathbb{R}.$$

For $0 \leq t \leq 1$ one has, more generally,

$$(11) \quad \lim_{x \rightarrow \pm\infty} F_{N,t}(x) = tc_N.$$

Proof. Let

$$S(u) = \sum_{n=1}^{\infty} \exp\left(\frac{u}{2} - \pi n^2 e^{2u}\right).$$

For $\vartheta(x) = \sum_{k \in \mathbb{Z}} e^{-\pi k^2 x}$, Jacobi's identity $\vartheta(x) = x^{-1/2} \vartheta(1/x)$ gives

$$S(u) - S(-u) = -\sinh\left(\frac{u}{2}\right).$$

Differentiating at $u = 0$ gives $S'(0) = -1/4$, hence

$$\frac{1}{2} + \sum_{n=1}^{\infty} (1 - 4\pi n^2) e^{-\pi n^2} = 0.$$

Subtracting (7) proves the tail formula in (9). Every tail summand is positive. The strict upper bound follows directly from (7), since every term in its finite sum is negative.

Finally, $\Phi_N \in L^1(0, \infty)$. The Riemann–Lebesgue lemma applied to (6) proves (10). $\square$

The shifted functions are not constant. Indeed, at $z = i/2$ one has $s = 0$, and the factor $s(s-1)$ in (1) vanishes. Since $\Xi_N = H_N + c_N$,

$$(12) \quad F_{N,t}(i/2) = \frac{1}{2} - (1-t)c_N, \quad F_{N,t}(i/2) - tc_N = \frac{1}{2} - c_N > 0.$$

4. GROWTH

Lemma 4.1. *For each fixed finite N and $t \in [0, 1]$, the entire function $F_{N,t}$ has order at most one. More precisely, if $M_{N,t}(r) = \max_{|z| \leq r} |F_{N,t}(z)|$, then*

$$\log M_{N,t}(r) = O_N(r \log(2+r)).$$

Proof. For $|z| \leq r$ and $u \geq 0$, $|\cos(zu)| \leq e^{ru}$. Since

$$0 < S_N(u) \leq N \exp\left(\frac{u}{2} - \pi e^{2u}\right),$$

the change of variables $t = e^{2u}$ gives

$$\begin{aligned} |I_N(z)| &\leq \frac{N}{2} \int_1^\infty t^{r/2-3/4} e^{-\pi t} dt \\ &\leq \frac{N}{2} \pi^{-(r/2+1/4)} \Gamma\left(\frac{r}{2} + \frac{1}{4}\right) \end{aligned}$$

for all sufficiently large r . Stirling's estimate, together with the quadratic factor in (4), yields the asserted bound. The identity $F_{N,t} = \Xi_N - (1-t)c_N$ shows that adding the bounded constant does not change the estimate. Its order consequence is immediate from the definition of the order of an entire function. $\square$

5. NONREAL ZEROS AT EVERY FINITE LEVEL

Proof of Theorem 1.1. Fix a finite $N \geq 1$ and $0 < t \leq 1$, and suppose that every zero of $F_{N,t}$ is real. Proposition 3.1 and (11) show that $F_{N,t}(x) > 0$ for all sufficiently large $|x|$. Equation (12) shows that $F_{N,t}$ is not the zero function, so its zeros are isolated. Under the supposition, all zeros therefore lie in a compact real interval and are finite in number.

By Lemma 4.1 and Hadamard factorization [1, Ch. 2],

$$(13) \quad F_{N,t}(z) = e^{az+b}P(z)$$

for constants $a, b \in \mathbb{C}$ and a polynomial P . Evenness gives the reflection product

$$Q(z) = F_{N,t}(z)F_{N,t}(-z) = e^{2b}P(z)P(-z)$$

is a polynomial. If $L = tc_N$, then (11) gives $Q(x) \rightarrow L^2$ as $x \rightarrow +\infty$. A polynomial with a finite limit on the positive real axis is constant, so $Q \equiv L^2$. Evenness now gives $F_{N,t}^2 \equiv L^2$. Since $L > 0$, the function $F_{N,t}$ never vanishes; differentiating shows that $F'_{N,t} \equiv 0$. Hence $F_{N,t}$ is constant, contrary to (12). Thus $F_{N,t}$ has a nonreal zero. Because $F_{N,t}$ is real and even, its nonreal zeros occur in the corresponding conjugation and sign symmetry orbit. $\square$

At $t = 1$, $F_{N,1} = \Xi_N$ is the direct incomplete-gamma truncation. Hence every finite truncation, including $N = 1$, has a nonreal zero. The same conclusion holds for every positive shift $0 < t \leq 1$.

Remark 5.1 (Local zero motion). Fix N . If z_0 is a simple zero of F_{N,t_0} for some $0 < t_0 \leq 1$, the holomorphic implicit-function theorem gives a unique local zero branch $z(t)$ through (t_0, z_0) , with

$$z'(t_0) = -\frac{c_N}{H'_N(z_0)}.$$

This identity is local: it gives no global ordering, collision exclusion, or continuation through a multiple zero.

6. RELATION TO EARLIER APPROXIMANTS AND SCOPE

For a single summand $S_n(u) = \exp(u/2 - \pi n^2 e^{2u})$, direct differentiation gives

$$(14) \quad 2 \left(S''_n(u) - \frac{1}{4} S_n(u) \right) = e^{-\pi n^2 e^{2u}} \left(8\pi^2 n^4 e^{9u/2} - 12\pi n^2 e^{5u/2} \right).$$

Consequently, (8) is exactly the normalization of the finite Fourier sums in Haglund's equations (3), (13), and (14) [3]. The direct incomplete-gamma truncation differs from that sum by the positive constant:

$$(15) \quad \Xi_N(z) = H_N(z) + c_N.$$

At the unshifted endpoint, Pólya proved that every finite partial sum of Riemann's kernel series has infinitely many zeros but only finitely many real zeros [5, §4]. Thus $H_N = F_{N,0}$ already has nonreal zeros. Theorem 1.1 covers every positive shift $0 < t \leq 1$; its boundary argument requires $tc_N > 0$ and therefore does not apply at $t = 0$.

Haglund investigated monotonic patterns among complex zeros and reported numerical zeros without rigorous error bounds [3]. His numbered conjectures concern the unshifted sums and a next-summand homotopy, not the additive constant path $H_N + tc_N$. Hejhal's work concerns other Pólya-type kernel modifications [4]. Neither result addresses the additive path $H_N + tc_N$, and no monotonicity statement follows from the local formula in Remark 5.1.

The conclusion is limited to the functions in (8) with $0 < t \leq 1$. It rules out the proof plan that demands real-rootedness for these finite approximants and then passes to Ξ by local uniform convergence. Changing the truncation, adding a nonconstant correction,

or proving a different source-specific zero theorem produces a different problem. The existence of nonreal zeros of the approximants has no implication for whether the zeros of the limiting Riemann Xi function are real.

DATA AND CODE AVAILABILITY

The proof is analytic and uses no external data or numerical zero location. An accompanying Lean 4 formalization checks the finite definitions, theta identity, boundary limits, nonconstancy witness, growth bound, factorization contradiction, symmetry orbit, and local-uniform convergence to the classical completed Xi function, together with the simple-zero local motion stated in Remark 5.1.

ASSISTANCE DISCLOSURE

AI systems were used extensively in mathematical derivation, Lean proof development, literature discovery, organization, and typesetting. The author remains responsible for every mathematical statement, proof, citation, and submission decision.

REFERENCES

1. Ralph Philip Boas, Jr., *Entire functions*, Pure and Applied Mathematics, vol. 5, Academic Press, New York, 1954, MR0068627.
2. Harold M. Edwards, *Riemann's zeta function*, Dover Publications, Mineola, NY, 2001, Reprint of the 1974 original; MR1854455; ISBN 978-0-486-41740-0.
3. James Haglund, *Some conjectures on the zeros of approximates to the Riemann Xi-function and incomplete gamma functions*, Central European Journal of Mathematics **9** (2011), no. 2, 302–318, [doi:10.2478/s11533-010-0095-3](https://doi.org/10.2478/s11533-010-0095-3); [arXiv:0910.5228](https://arxiv.org/abs/0910.5228).
4. Dennis A. Hejhal, *On a result of G. Pólya concerning the Riemann ξ -function*, Journal d'Analyse Mathématique **55** (1990), no. 1, 59–95, [doi:10.1007/BF02789198](https://doi.org/10.1007/BF02789198); MR1094717.
5. George Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ -Funktion*, Acta Mathematica **48** (1926), 305–317, [doi:10.1007/BF02565336](https://doi.org/10.1007/BF02565336); [English translation](#).